

CANON PERMUTATIONS AVOIDING 213 AND 312 ARE COUNTED BY FOUR CATALAN NUMBERS

ABSTRACT. Let C_3^k be the set of canon permutations of the multiset $\{1^k, 2^k, 3^k\}$, and let $c_3^k(\Lambda)$ count those avoiding every pattern in Λ . Laudone conjectured that $c_3^k(213, 312) = 4C_k$ for all $k \geq 1$, where C_k is the k -th Catalan number. We prove it, in the refined form $c_3^{k,\sigma}(213, 312) = C_k$ for each of the four canon witnesses 123, 132, 231, 321, and 0 for the other two, by exhibiting the surviving permutations explicitly: each is a solid block 1^k at one end of the word and a ballot word on the letters 2 and 3 at the other; the two choices of end and the two choices of which of 2, 3 dominates the ballot condition are what make four families of C_k words and not two. The proof is a six-case analysis of the canon witness and uses the hypothesis $n = 3$ in exactly one place, namely that over a three-letter alphabet an occurrence of a three-letter pattern is a literal subsequence. The same observation shows that relabelling the alphabet is a bijection of C_3^k transporting avoidance, whence $c_3^k(123, 321)$ and $c_3^k(132, 231)$ equal $4C_k$ as well, for every k ; and that $c_3^k(\{\sigma\})$ does not depend on which $\sigma \in S_3$ is forbidden. The cells $k = 1$ and $k = 2$ of the conjecture are already in print.

1. THE CONJECTURE

For $n, k \geq 1$ let S_n^k be the set of permutations of the multiset $\{1^k, 2^k, \dots, n^k\}$: the words of length nk that contain each letter of $[n]$ exactly k times. For a letter a and $j \in [k]$ write $\text{pos}_a(j)$ for the position of the j -th occurrence of a . A word $\pi \in S_n^k$ is *canon* if there is a permutation $\sigma \in S_n$ such that, for every $j \in [k]$, the subsequence of j -th copies of the letters read left to right spells σ ; equivalently

$$(1) \quad \text{pos}_{\sigma_1}(j) < \text{pos}_{\sigma_2}(j) < \dots < \text{pos}_{\sigma_n}(j) \quad \text{for all } j \in [k].$$

Reading (1) at $j = 1$ shows that σ is the order in which the letters first occur, so σ is determined by π ; we call it the *witness* of π and write C_n^k for the set of canon permutations and $C_n^{k,\sigma}$ for those with witness σ . As usual π *contains* the pattern $p \in S_m$ if some length- m subsequence of π has the same relative order as p , and *avoids* p otherwise; for a set $\Lambda \subseteq S_m$ we put

$$\begin{aligned} c_n^k(\Lambda) &= \#\{\pi \in C_n^k : \pi \text{ avoids every } p \in \Lambda\}, \\ c_n^{k,\sigma}(\Lambda) &= \#\{\pi \in C_n^{k,\sigma} : \pi \text{ avoids every } p \in \Lambda\}. \end{aligned}$$

2020 *Mathematics Subject Classification*. Primary 05A05; Secondary 05A15, 05A19, 05E10.

Key words and phrases. canon permutation, pattern avoidance, Catalan number, ballot sequence, standard Young tableau, multiset permutation.

We write $C_k = \frac{1}{k+1} \binom{2k}{k}$, so $C_1, C_2, C_3, \dots = 1, 2, 5, 14, 42, 132, \dots$; this is the indexing the source itself uses.

Laudone [2] asks (his Question 6.5) for $c_3^k(\Lambda)$ as Λ ranges over subsets of S_3 , with $n = 3$ fixed and k varying, and offers one instance as an example; it is the last mathematical statement of that paper, and we quote it verbatim. It appears in the final section, and is numbered Conjecture 6.6 in the compiled source.

Conjecture (Laudone). For $k \geq 1$, $c_3^k(213, 312) = 4C_k$ where C_k is the k -th Catalan number.

Our results are the following. Theorem 1 proves the conjecture and refines it by the witness; Theorem 2 shows that the same argument settles two further pattern pairs, and Corollary 3 one further family.

Theorem 1. For every $k \geq 1$,

$$c_3^{k,\sigma}(213, 312) = \begin{cases} C_k, & \sigma \in \{123, 132, 231, 321\}, \\ 0, & \sigma \in \{213, 312\}, \end{cases}$$

and consequently $c_3^k(213, 312) = 4C_k$. Moreover the survivors are exactly the following four families of words, where u always denotes a word on k letters 2 and k letters 3:

- (i) $1^k u$ with the j -th 2 of u before its j -th 3 for all j (these have witness 123);
- (ii) $1^k u$ with the j -th 3 before the j -th 2 (witness 132);
- (iii) $u 1^k$ with the j -th 2 before the j -th 3 (witness 231);
- (iv) $u 1^k$ with the j -th 3 before the j -th 2 (witness 321).

Theorem 2. For every $\tau \in S_3$ and every $\Lambda \subseteq S_3$ we have $c_3^k(\Lambda) = c_3^k(\tau\Lambda)$, where $\tau\Lambda = \{\tau \circ p : p \in \Lambda\}$. The orbit of $\{213, 312\}$ under this action is

$$\{ \{213, 312\}, \{123, 321\}, \{132, 231\} \},$$

so for every $k \geq 1$

$$c_3^k(213, 312) = c_3^k(123, 321) = c_3^k(132, 231) = 4C_k.$$

Corollary 3. $c_3^k(\{\sigma\})$ is the same for all six $\sigma \in S_3$.

The cells $k = 1$ and $k = 2$ of the conjecture are already in print. At $k = 1$ it says that four of the six permutations of S_3 avoid 213 and 312. At $k = 2$ the canon permutations are precisely the nonnesting permutations of Elizalde and Luo [1], as [2] itself remarks, and their theorem $c_n^2(132, 231) = 2^n$ gives $c_3^2(132, 231) = 8$; the complementation $x \mapsto 4 - x$ is the relabelling $\tau = 321$ of Theorem 2, which carries $\{213, 312\}$ to $\{132, 231\}$, so $c_3^2(213, 312) = 8 = 4C_2$ follows from [1]. The argument below is elementary and short; we make no claim of difficulty.

2. CONTAINMENT OVER THREE LETTERS IS LITERAL

Everything below rests on one observation, and it is the only place where the hypothesis $n = 3$ enters.

Lemma 4. *Let w be a word over the alphabet $\{1, 2, 3\}$ and let $p \in S_3$. Then w contains p if and only if $p_1 p_2 p_3$ occurs in w as a literal subsequence, that is, if and only if there are positions $i < j < l$ with $w_i = p_1$, $w_j = p_2$, $w_l = p_3$.*

Proof. An occurrence of p is a triple of positions $i < j < l$ whose values $(a, b, c) = (w_i, w_j, w_l)$ have the same relative order as p . Since p has three distinct entries, a, b, c are pairwise distinct; being three distinct elements of $\{1, 2, 3\}$ they are 1, 2, 3 in some order, and having the same relative order as p then forces $(a, b, c) = (p_1, p_2, p_3)$. The converse is immediate. $\square$

In particular, over $\{1, 2, 3\}$: w contains 213 iff some 2 precedes some 1 which precedes some 3, and w contains 312 iff some 3 precedes some 1 which precedes some 2. We use these two sentences without further comment.

Remark 5. Lemma 4 is false for larger alphabets, and 314 is a witness: it contains 213 (the values 3, 1, 4 have the relative order of 2, 1, 3) but has no literal 2, 1, 3. This is exactly why nothing in this note says anything about $c_n^k(213, 312)$ for $n \geq 4$.

3. PROOF OF THEOREM 1

Fix $k \geq 1$ and write $\Lambda = \{213, 312\}$. Since the witness is determined by the word, the six sets $C_3^{k, \sigma}$ partition C_3^k , and it suffices to treat each witness.

(a) $\sigma = 213$ and $\sigma = 312$ are empty. If $\sigma = 213$ then (1) at $j = 1$ gives $\text{pos}_2(1) < \text{pos}_1(1) < \text{pos}_3(1)$, so the letters 2, 1, 3 occur in that order and π contains 213 by Lemma 4. If $\sigma = 312$ then $\text{pos}_3(1) < \text{pos}_1(1) < \text{pos}_2(1)$ and π contains 312. Hence $c_3^{k, 213}(\Lambda) = c_3^{k, 312}(\Lambda) = 0$.

(b) $\sigma = 123$. Let $\pi \in C_3^{k, 123}$ avoid Λ . We claim all k letters 1 come first. If not, some letter of $\{2, 3\}$ precedes some 1; let q be the leftmost position of a 1 having a non-1 to its left, say $q = \text{pos}_1(j)$.

- If a 2 occurs before q : by (1), $\text{pos}_3(j) > \text{pos}_2(j) > \text{pos}_1(j) = q$, so some 3 occurs after q ; that 2, the 1 at q and that 3 give an occurrence of 213, a contradiction.
- If no 2 but some 3 occurs before q : then $\text{pos}_3(1) < q < \text{pos}_2(1)$, contradicting $\text{pos}_2(1) < \text{pos}_3(1)$, which is (1) for $\sigma = 123$.

So $\pi = 1^k u$ with u a word on k letters 2 and k letters 3. For such a word $\text{pos}_1(j) = j$ is smaller than every position of u , so (1) reduces to $\text{pos}_2(j) < \text{pos}_3(j)$, i.e. the j -th 2 of u precedes its j -th 3, for every j . Conversely every such word avoids Λ : an occurrence of 213 or of 312 needs a 1 strictly to the right of a 2 or of a 3, and there is none. Reading 2 as an up-step and 3 as a down-step identifies these u with the Dyck paths of semilength k , since “the j -th 2 precedes the j -th 3 for all j ” says exactly that every prefix of u has

at least as many 2's as 3's; the Catalan count of those paths is standard, see [3]. Hence $c_3^{k,123}(\Lambda) = C_k$, realised by family (i).

(c) $\sigma = 231$. Let $\pi \in C_3^{k,231}$ avoid Λ and let $q = \text{pos}_1(1)$. By (1), $\text{pos}_2(1) < \text{pos}_3(1) < q$, so both a 2 and a 3 occur before q . If some 3 occurred after q we would have $2 \cdots 1 \cdots 3$ and hence the pattern 213; if some 2 occurred after q we would have $3 \cdots 1 \cdots 2$ and hence 312. So every position after q carries a 1, i.e. $\pi = u 1^k$ with u a word on k letters 2 and k letters 3. Now $\text{pos}_1(j) = 2k + j$ exceeds every position of u , so (1) again reduces to “the j -th 2 of u precedes its j -th 3 for every j ”, and conversely each such word avoids Λ because no 2 or 3 follows a 1. Hence $c_3^{k,231}(\Lambda) = C_k$, realised by family (iii).

(d) $\sigma = 132$ and $\sigma = 321$. Let t be the relabelling that exchanges the letters 2 and 3 and fixes 1. Applying t letterwise is an involution of S_3^k ; by (1) it carries $C_3^{k,\sigma}$ onto $C_3^{k,t\circ\sigma}$, and by Lemma 4 it carries occurrences of 213 to occurrences of 312 and back. Since t therefore permutes Λ , it restricts to a bijection between the Λ -avoiders with witness σ and those with witness $t \circ \sigma$. As $t \circ 123 = 132$ and $t \circ 231 = 321$, cases (b) and (c) give $c_3^{k,132}(\Lambda) = c_3^{k,321}(\Lambda) = C_k$, and applying t to families (i) and (iii) yields (ii) and (iv).

Summing, $c_3^k(\Lambda) = 0 + 0 + 4C_k = 4C_k$. $\square$

4. PROOF OF THEOREM 2 AND COROLLARY 3

Let $\tau \in S_3$ act on a word letterwise. This is a bijection of S_3^k ; by (1) a canon word with witness σ is sent to a canon word with witness $\tau \circ \sigma$, so τ is a bijection of C_3^k . By Lemma 4, containment of $p \in S_3$ is containment of the literal word $p_1 p_2 p_3$, and τ carries a literal occurrence of $p_1 p_2 p_3$ to a literal occurrence of $\tau(p_1) \tau(p_2) \tau(p_3)$ and back. Hence π avoids Λ if and only if $\tau(\pi)$ avoids $\tau\Lambda$, and $c_3^k(\Lambda) = c_3^k(\tau\Lambda)$.

Writing τ as the word $\tau(1)\tau(2)\tau(3)$, one computes $\tau\{213, 312\}$ for the six τ and finds $\{213, 312\}$ for $\tau \in \{123, 132\}$, $\{123, 321\}$ for $\tau \in \{213, 231\}$ and $\{132, 231\}$ for $\tau \in \{312, 321\}$. So the stabiliser is the order-two group generated by the transposition of 2 and 3 and the orbit has size three, as stated; Theorem 1 then gives the common value $4C_k$.

For Corollary 3, S_3 acts simply transitively on the singletons $\{\sigma\}$, since $\tau \circ p = p'$ has the unique solution $\tau = p'p^{-1}$. Hence all six values $c_3^k(\{\sigma\})$ coincide. $\square$

5. VERIFICATION

A companion program `verify.py` (Python 3.9+, standard library only, exact integer arithmetic) re-derives the quantities asserted here. It builds C_3^k twice — once from definition (1) over all of S_3^k and once from standard Young tableaux — decides containment twice, once by relative order and once literally, and confirms the refinement of Theorem 1 and the four families

by exhaustive enumeration for $k \leq 6$; a transfer-matrix recursion that uses only the prefix form of (1) and a greedy pattern matcher, and knows nothing of Theorem 1, reproduces the refinement and the value $4C_k$ for all $k \leq 24$ and, for the witness 123 alone, for all $k \leq 40$. It also verifies Lemma 4 exhaustively over the alphabet, exhibits its failure at $n = 4$, and checks Theorem 2 object by object for $k \leq 3$. No k beyond those ranges, no statement about $n \geq 4$ and no bibliographic assertion is machine-checked; the proof above is what carries the general case.

REFERENCES

- [1] S. Elizalde and A. Luo, *Pattern avoidance in nonnesting permutations*, Discrete Math. Theor. Comput. Sci. **27** (2025), no. 1 (Permutation Patterns 2024 special issue), published 17 October 2025; doi:10.46298/dmtcs.14885; arXiv:2412.00336.
- [2] R. Laudone, *Pattern avoidance in canon permutations*, preprint, 21 August 2026; arXiv:2608.21351.
- [3] R. P. Stanley, *Catalan Numbers*, Cambridge University Press, Cambridge, 2015.